Unit-2

Lecture-1

Step-Wise Planning

Step-wise planning means:

- Planning a software project in small stages instead of one big plan. This is important because software requirements change frequently.

Steps in Step-Wise Planning

Step 1 – Define Objectives

- Why are we developing this software?
- Example:
- To build an Online Attendance System
- To reduce manual work

Step 2 – Define Scope

- What features will the software have?
- Example:
- Login system
- Attendance marking
- Report generation
- Not included:
- Fee management
- Hostel management

Step 3 – Identify Deliverables

- What will be given to the customer?
- Requirement document
- Software
- User manual
- Test report

Step 4 – Break into Tasks (WBS- Work Breakdown Structure)

- Break big work into small jobs.

Step 5 – Schedule Tasks

- Decide:
- When each task starts
- When it ends
- Example:

Design – 1 week
Coding – 3 weeks
Testing – 2 weeks

Step 6 – Assign Resources

- Who will do the work?
- 2 programmers
- 1 tester
- 1 project manager

Step 7 – Review and Update

- After every stage:
- Check progress
- Change plan if needed

Lecture-2

PRINCE2 is a structured project management method

It means:

Projects IN Controlled Environments

It ensures

proper planning

Strong control

Clear responsibility

Principle	Meaning
Business justification	Project must be useful
Learn from experience	Use past mistakes
Defined roles	Everyone knows their job
Manage by stages	Work in phases
Manage by exception	Report only problems
Focus on products	Focus on outputs
Tailor	Adjust for project size

PRINCE2 Processes

- Starting the project
- Initiating the project
- Controlling the stages
- Managing product delivery
- Closing the project

- It is a structured project management methodology used worldwide for managing software, IT, construction, and government projects.

Its main goal is:

- To ensure that a project is well-planned, well-controlled, and delivers business value.

Why PRINCE2 is important in Software Projects

•Software projects often:

•Change requirements

•Have risks

•Overrun time and cost

PRINCE2 provides:

Control
Documentation
Responsibility
Risk management

So projects do not go out of control.

Lecture-3 PMBOK

- PMBOK stands for Project Management Body of Knowledge. It is the collection of processes like best practices, terminologies, and guidelines that are accepted as standards within the project management Industry. It is considered valuable for companies because it helps them standardize practices across various departments and tailor processes to suit specific needs.

- It constitutes a collection of project management processes, best practices, terminologies, guidelines, and tools, accepted as standard within the project management industry.
- It is widely used as a standard that allows organizations to professionalize the role of project management resulting in better-managed projects.

Why Use PMBOK?

- **Project Success:** To increase project success within the set timeframe and budget. PMBOK enables you to do precisely that by providing the best practices for project management.
- **Standardized Approach:** PMBOK allows companies to standardize practices across departments. So, it doesn't matter whether a project belongs to the development team or the operations team. All projects are managed in the same manner.

- **Standardized System with other companies:** Additionally, PMBOK can help project managers work with a standardized system across companies. If you leave your organization for another, you'll still be able to use the same practices described by PMBOK
- **Project Risk Analysis:** Think of PMBOK as a project management handbook that you can refer to whenever you need guidance. The methods documented within the PMBOK methodology can help you manage potential project risks. The guide discusses what works, giving you an idea of what you need to do to steer your projects towards success. It also discusses what doesn't work, giving you the power to prevent the failure of projects.

What can Project Managers do with PMBOK?

•Project managers can become more adaptable by becoming familiar with the standardised procedures, policies, and best practices used in project management by using PMBOK. You can also go to a new company or industry without having to start over if you are familiar with best practices in project management from scratch and domains.

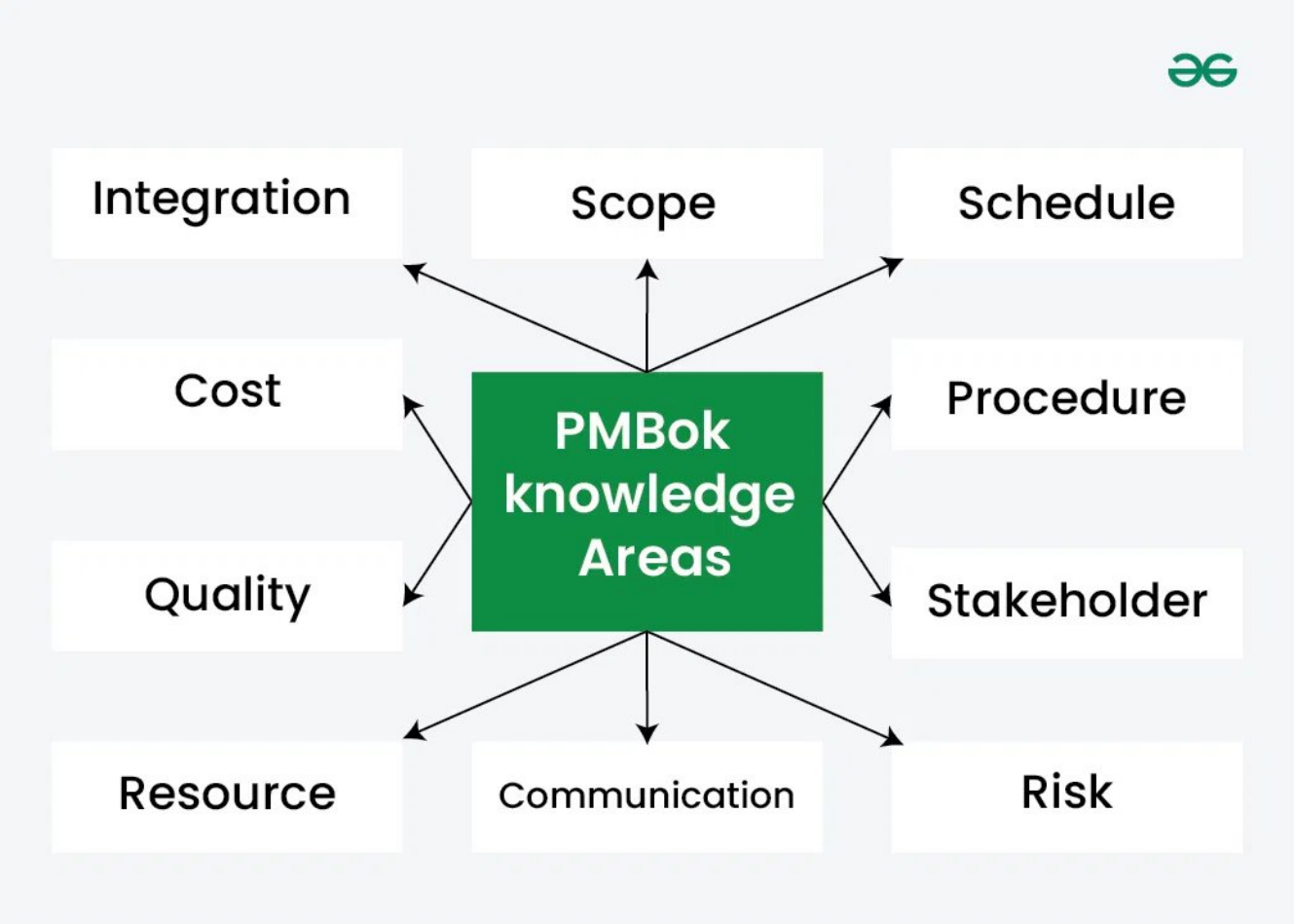
Here are some key actions project managers can take with PMBOK:

- **Framework Adoption**: Project managers can adopt the PMBOK framework as a guide for structuring their projects. This involves understanding the five process groups (Initiating, Planning, Executing, Monitoring and Controlling, and Closing) and the ten knowledge areas (Integration, Scope, Time, Cost, Quality, Human Resource, Communications, Risk, Procurement, and Stakeholder Management) outlined in PMBOK.

- **Standardized Terminology**: Ensure consistent communication by using the standardized terminology and definitions provided in PMBOK. This helps in aligning the understanding of project concepts among team members and stakeholders.
- **Project Planning**: Utilize PMBOK's guidelines and best practices for project planning. This involves developing project management plans, defining project scope, creating schedules, estimating costs, identifying risks, and planning communications and stakeholder engagement strategies.

- **Execution and Monitoring**: Implement project management processes and techniques outlined in PMBOK during project execution and monitoring phases. This includes coordinating project activities, managing resources, tracking progress, controlling costs, and assessing and responding to risks and issues.

PMBOK Knowledge Areas



Lecture-4 Cost-Benefit Analysis

- In product management, resources such as time, budget, and personnel are finite. Cost-benefit analysis provides a structured framework to assess whether a proposed initiative, such as a new feature, product launch, or process improvement, is worth pursuing.
- According to a 2024 PMI report, organizations that consistently use CBA for project prioritization achieve 20% higher ROI on product investments compared to those that don’t.

- Informed Decision-Making: Ensures decisions are grounded in data, not assumptions.
- Resource Optimization: Directs resources to high-impact initiatives.
- Stakeholder Alignment: Provides a clear rationale for prioritizing or deprioritizing projects.
- Risk Mitigation: Identifies potential financial or strategic risks early.

Steps to Conduct a Cost-Benefit Analysis

1. Define the Scope and Objectives

Clearly outline the initiative being evaluated (e.g., building a new feature, entering a new market, or improving infrastructure). Define the objectives, such as increasing user retention, boosting revenue, or reducing churn. For example, a product manager for a SaaS platform might evaluate the addition of an AI-powered analytics dashboard to enhance user engagement.

2. Identify Costs

- List all costs associated with the initiative, both direct and indirect. Categorize them for clarity:
- **Direct Costs**: Development expenses (e.g., engineering hours, software licenses), marketing costs, or hardware purchases.
- **Indirect Costs**: Opportunity costs (e.g., resources diverted from other projects), training, or maintenance.
- **One-Time Costs**: Initial investments like design or prototyping.
- **Ongoing Costs**: Recurring expenses like server costs or customer support.

3. Identify Benefits

- Quantify the expected benefits, which may be financial, strategic, or user-focused. Common benefits include:
- **Revenue Gains**: Increased subscriptions, upsells, or new customer acquisition.
- **Cost Savings**: Reduced support tickets or operational efficiencies.
- **User Benefits**: Improved retention, engagement, or satisfaction (e.g., higher Net Promoter Score).
- **Strategic Advantages**: Competitive differentiation or market expansion.

4. Quantify Costs and Benefits

- Assign monetary values to costs and benefits where possible. For non-monetary benefits (e.g., user satisfaction), use proxies like estimated revenue impact or industry benchmarks. Consider the time horizon (e.g., 1-3 years) to account for long-term benefits versus upfront costs. Use tools like discounted cash flow (DCF) to calculate the present value of future benefits, especially for long-term projects.
- **For example:**
- **Costs**: $60,000 one-time + $5,000/month ongoing = $120,000 over 2 years.
- **Benefits**: $120,000/year revenue increase = $240,000 over 2 years.

5. Compare Costs and Benefits

- Calculate the net benefit by subtracting total costs from total benefits:
- **Net Benefit** = Total Benefits ($240,000) – Total Costs ($120,000) = $120,000.
- Alternatively, use metrics like:
- **Return on Investment (ROI)**: (Net Benefit / Total Costs) × 100 = ($120,000 / $120,000) × 100 = 100%.
- **Payback Period**: Time to recover costs (e.g., $120,000 / $120,000/year = 1 year).
- **Benefit-Cost Ratio (BCR)**: Total Benefits / Total Costs = $240,000 / $120,000 = 2.0 (a ratio > 1 indicates positive value).

6. Assess Risks and Sensitivities

- Consider risks that could impact costs or benefits, such as:
- **Technical Risks**: Development delays or bugs.
- **Market Risks**: Lower-than-expected user adoption.
- **External Risks**: Regulatory changes or competitor actions.
- Perform a sensitivity analysis by adjusting key assumptions (e.g., what if revenue increases only 5% instead of 10%?). This helps evaluate the initiative's robustness under different scenarios.

7. Make a Recommendation

Based on the analysis, recommend whether to proceed, modify, or abandon the initiative. Present findings to stakeholders with clear visuals (e.g., charts or tables) and a concise summary of costs, benefits, risks, and ROI. For the AI dashboard, a positive ROI and short payback period might justify moving forward, provided risks are manageable.

Tools for Cost-Benefit Analysis

•Leverage tools to streamline the process:

•**Spreadsheets**: Excel or Google Sheets for cost-benefit calculations and visualizations.

•**Analytics Platforms**: Mixpanel or Amplitude to estimate user-driven benefits like retention.

•**Survey Tools**: Typeform or Qualtrics to gather user feedback for quantifying benefits.

•**Project Management Tools**: Jira or Asana to estimate development costs.

•**Financial Modeling**: Tools like Tableau for advanced ROI or DCF analysis.